#4 That's not enough, though: we also need to check that the graph is bipartite and the two partitions have the right size.

Listing 15.5, instead, shows the method to check if a graph is (isomorphic to) $K_{3,3}$: this is a bit more complicated because it's not enough to check that the count of vertices (6) and edges (9) is right: we also need to check that the graph is bipartite and that both partitions have size 3.

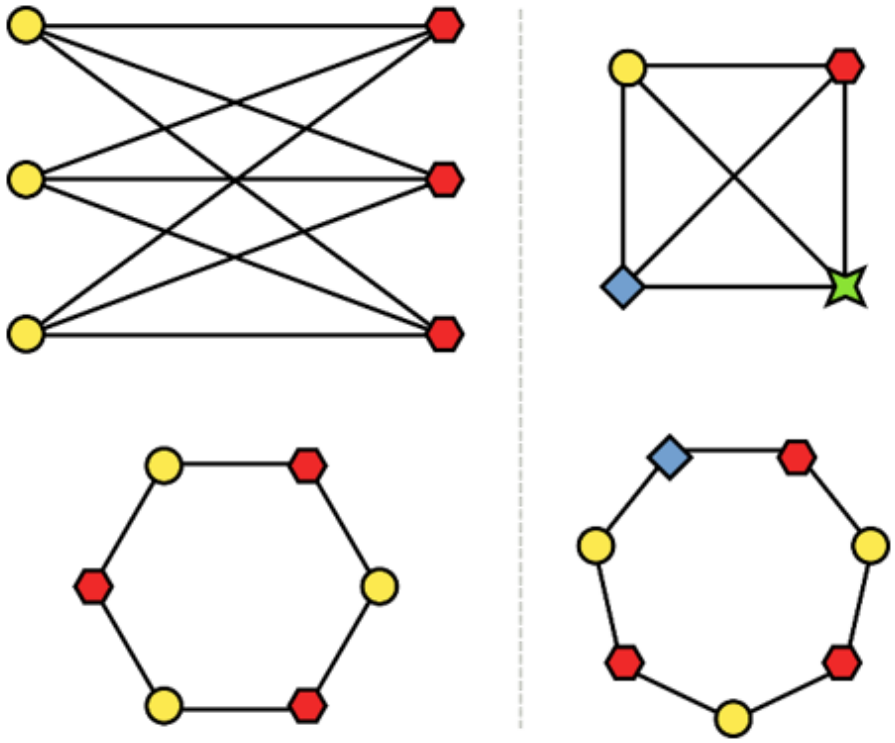


Figure 15.10 Bipartite graphs and coloring. All graphs are colored with the minimum possible number of colors. On the left, two examples of bipartite graphs; on the right, non-bipartite graphs: as you can see, two colors are not enough for these graphs (we also use different shapes, along with colors, for the vertices to help visualizing the difference).